

Chapter-11

Representation & Description

Chain Code

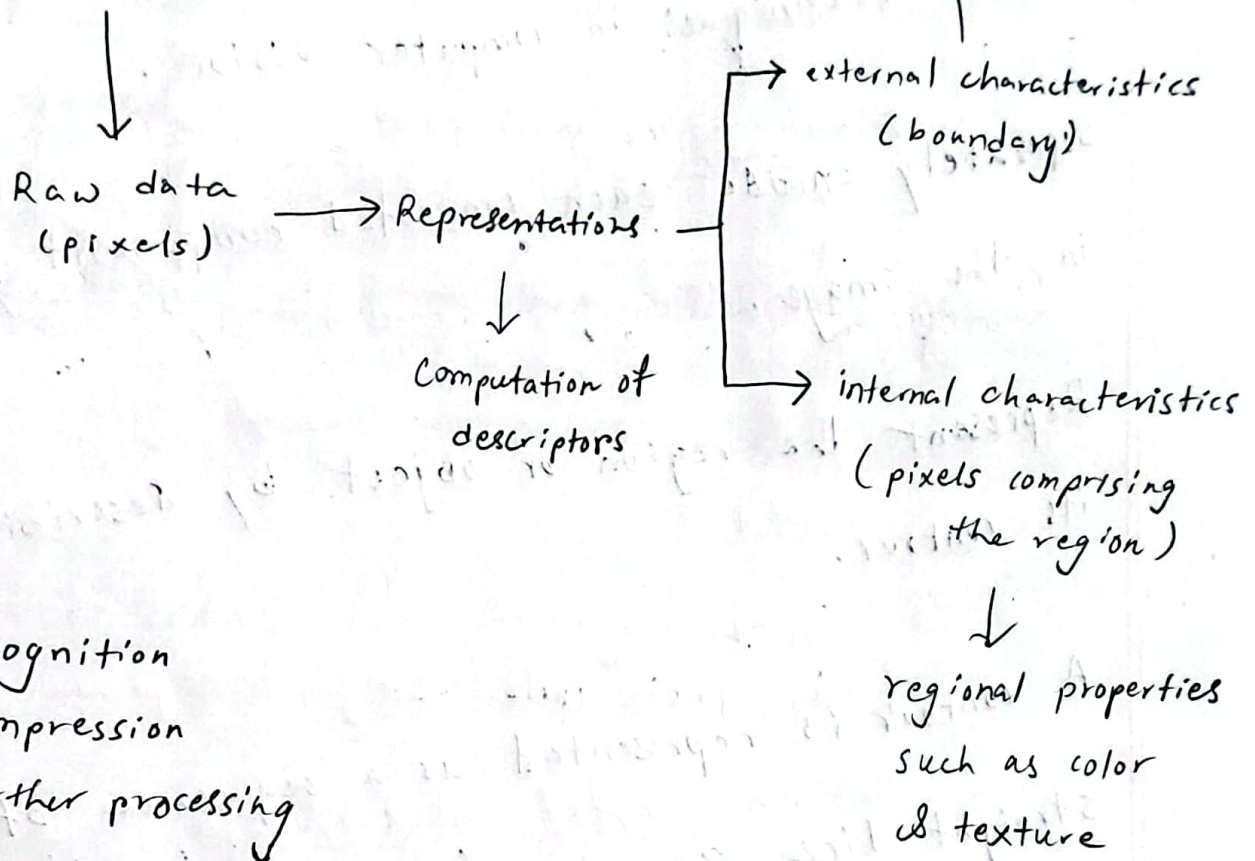
Image representation

Represent and describe information embedded in an image in other forms that are more suitable than the image itself.

It is easier to understand, requires fewer memory and faster to be processed. It is more ready to be used.

Image processing

Segmentation



Uses

- (i) recognition
- (ii) compression
- (iii) further processing

NOTE

Must ensure →

- (i) Translation invariance
- (ii) Scale invariance
- (iii) Rotation invariance

Chain Code

Freeman chain codes are actually one of the oldest techniques in computer vision.

Seperately encode each connected component / blob in the image.

Represent the region or object by describing its contour.

A contour is represented as a sequence of straight line segments of specified length and direction.

Types of chain code

8-neighbour $\rightarrow$ It will follow the contour and keep generating direction number that is generated in anti-clockwise manner.

4-neighbour $\rightarrow$ Same but with 4 neighbour.

It is also known as crack code.

chain codes for 8-neighbour

3	2	1
4	x	0
5	6	7

chain codes for 4-neighbour =

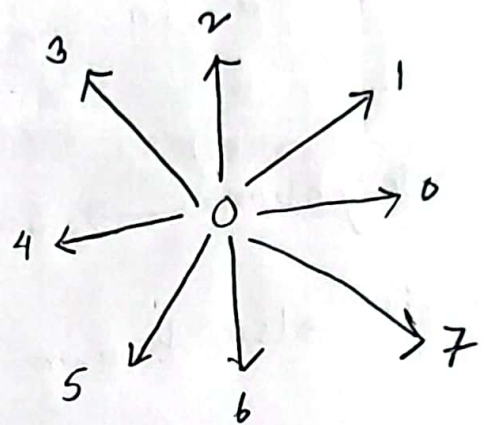
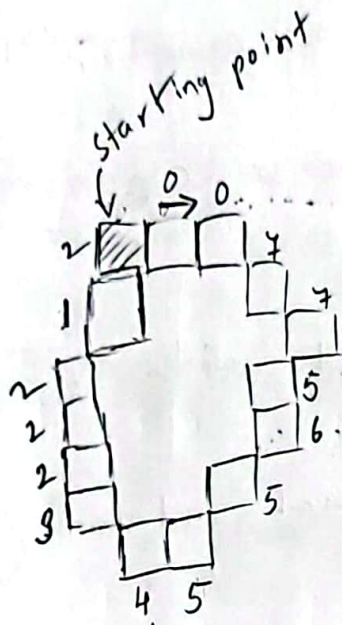
2	x	0
3		

Note

Even codes $\{0, 2, 4, 6\}$ correspond to horizontal and vertical directions.

Odd codes $\{1, 3, 5, 7\}$ correspond to the diagonal directions.

Ex-1



find chain code using 8-neighbour

Answer $\rightarrow 0, 0, 7, 7, 5, 6, 5, 5, 4, 3, 2, 2, 2, 1, 2$

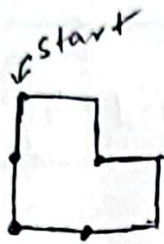
Advantages

Boundary reduced to a sequence of numbers.

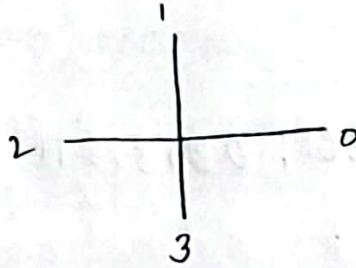
Provides good compression of boundary description.

They are translation invariant.

Ex-2

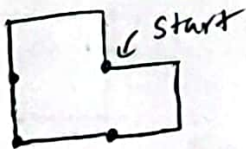


find chain code using 4-neighbour



CC $\rightarrow$ 03032211

Ex-3



find chain code using 4-neighbour

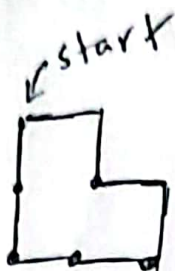
CC $\rightarrow$ 03221103

NOTE

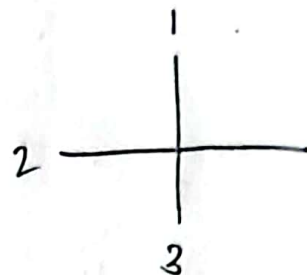
Chain code changes with a change in starting point, reflection & rotation.

Circular first difference & Normalized CFD

Ex-1



CC : 03032211

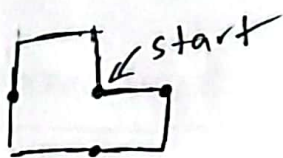


find CFD & NCFD

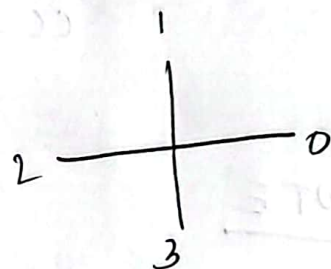
CFD : 31330303

NCFD : 03033133

Ex-2



CC : 03221103

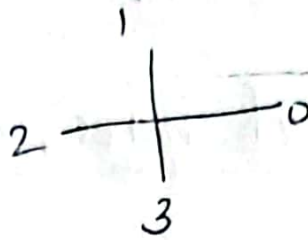
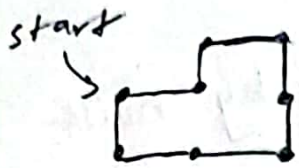


find CFD & NCFD

CFD : 33030331

NCFD : 03033133

Ex-3



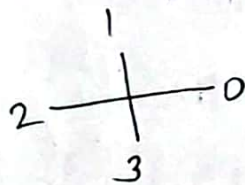
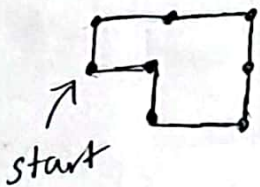
Find CC, CFD, NCFD

CC : 0 1 0 3 3 2 2 1

CFD : 1 3 3 0 3 0 3 3

NCFD : 0 3 0 3 3 1 3 3

Ex-4



Find CC, CFD, NCFD

CC : 1 0 0 3 3 2 1 2

CFD : 3 0 3 0 3 3 1 3

NCFD : 0 3 0 3 3 1 3 3

NOTE

Long chain of codes, easily disturbed by noise and sidetrack. It can be solved by resampling using larger grid spacing. Another issue was starting point but normalization will fix it.

————— END of chapter-11 —————